

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing
and Big Data
Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I, M.V. Rakesh Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.V. Rakesh Reddy

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.
 Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
 Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
 Some applications must remain on-premises due to compliance requirements.
 Design a hybrid architecture that ensures secure communication between environments.
 Explain how identity management and encryption should be implemented.
 Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
 The system must continue operating even if one region goes down.
 Propose a multi-region deployment strategy in the cloud.
 Explain how data replication and failover mechanisms should be configured.
 Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively Decisions are	Applies concepts with minor mistakes Reasonable	Partially correct application	Weak application
Decision Justification	20%	well justified and technically strong Clear and well-	justification	Weak justification	No useful justification
Clarity	15%	organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Architecture for Video Streaming Platform

Task Description: Design a cloud architecture with scalability and caching techniques for a media company handling millions of concurrent streaming users with minimal buffering, and explain load balancing, auto-scaling, and cost optimization strategies[cite: 3].

Cloud Architecture Design & Caching:

- Deploy a multi-tier architecture utilizing globally distributed Content Delivery Networks (CDNs) to cache video chunks at edge locations close to viewers[cite: 3].
- Store master video assets in highly durable cloud object storage, utilizing trans-coding pipelines to serve adaptive bitrate streaming formats[cite: 3].

Load Balancing and Auto-Scaling:

- Implement global and regional application load balancers to distribute incoming streaming requests evenly across backend media servers, preventing bottlenecks[cite: 3].
- Configure horizontal auto-scaling rules based on network throughput and CPU metrics to spin up compute instances dynamically during peak viewership hours[cite: 3].

Cost Optimization Strategies:

- Utilize reserved instances or Savings Plans for baseline compute workloads, combined with spot instances for elastic video encoding tasks[cite: 3].
- Implement intelligent data lifecycle policies to tier older or less-accessed video content into cheaper cold storage tiers[cite: 3].

2. Multi-Cloud Container Orchestration for Fintech Startup

Task Description: Propose a solution using container orchestration and multi-cloud strategies to solve deployment consistency challenges for a fintech startup, explaining service discovery, scaling, benefits, and risks[cite: 3].

Orchestration and Multi-Cloud Strategy:

- Adopt Kubernetes as the universal container orchestration layer across multiple cloud providers (e.g., AWS, Azure, GCP) to ensure declarative deployments and environment consistency[cite: 3].
- Use infrastructure-as-code (IaC) tools like Terraform to maintain uniform cluster provisioning and configuration management[cite: 3].

Service Discovery and Scaling:

- Leverage native Kubernetes service discovery, DNS management, and ingress controllers to route internal microservices traffic securely[cite: 3].
- Employ Horizontal Pod Autoscalers (HPA) and Cluster Autoscalers to scale application replicas automatically based on transaction volume and resource utilization[cite: 3].

Benefits and Risks:

- **Benefits:** Eliminates vendor lock-in, enhances high availability, and provides geographic resilience[cite: 3].
- **Risks:** Increases operational complexity, networking overhead across clouds, and data synchronization hurdles[cite: 3].

3. Backup and Disaster Recovery for E-Learning Platform

Task Description: Design a cloud-native backup, replication, and versioning solution to prevent data loss from accidental deletion, and explain how RTO, RPO, and business continuity are achieved[cite: 3].

Cloud-Native Backup and Versioning Solution:

- Enable object versioning and MFA-delete safeguards on cloud storage buckets containing course materials and user data to protect against accidental deletions[cite: 3].
- Configure automated snapshot policies for databases and block storage volumes, replicating them asynchronously to secondary disaster recovery regions[cite: 3].

Achieving RTO and RPO:

- **Recovery Point Objective (RPO):** Minimized to near-zero through continuous real-time data replication and frequent transactional snapshots[cite: 3].
- **Recovery Time Objective (RTO):** Minimized by maintaining pre-configured infrastructure templates and automated failover scripts ready for immediate execution[cite: 3].

Steps for Business Continuity:

- Establish routine automated recovery drills and data integrity validation tests[cite: 3].
- Deploy automated DNS failover routing policies to switch user traffic to standby environments seamlessly during system anomalies[cite: 3].

4. Hybrid Cloud Model for Government Agency

Task Description: Design a hybrid cloud architecture ensuring secure communication between on-premises compliance-bound workloads and cloud environments, detailing identity management, encryption, and management challenges[cite: 3].

Hybrid Architecture Design:

- Keep sensitive citizen records and core databases on-premises to meet strict compliance mandates, while deploying scalable public-facing portals in the cloud[cite: 3].
- Establish a secure communication bridge between environments utilizing dedicated private interconnects (e.g., AWS Direct Connect / Azure ExpressRoute) paired with encrypted IPsec VPN tunnels[cite: 3].

Identity Management and Encryption:

- Implement centralized identity and access management through federated single sign-on (SSO) and Active Directory integration across environments[cite: 3].
- Enforce rigorous encryption standards, securing data in transit via TLS and data at rest using agency-managed encryption keys[cite: 3].

Challenges in Management:

- Managing complex network routing policies, maintaining consistent security monitoring across disparate infrastructures, and troubleshooting latency issues[cite: 3].

5. Multi-Region High Availability Strategy for SaaS Provider

Task Description: Propose a multi-region deployment strategy for high availability and fault tolerance, explaining data replication, failover mechanisms, and proactive monitoring for a SaaS provider[cite: 3].

Multi-Region Deployment Strategy:

- Deploy active-passive or active-active application clusters across two or more distinct geographic cloud regions to ensure uninterrupted service even during catastrophic regional outages[cite: 3].

Data Replication and Failover Mechanisms:

- Configure cross-region database replication with low-latency synchronization to ensure data parity across regions[cite: 3].
- Utilize latency-based or failover routing via cloud DNS services to detect region unresponsiveness and automatically redirect user traffic to a healthy secondary region[cite: 3].

Monitoring and Alerting Strategies:

- Implement comprehensive distributed tracing, synthetic user transaction testing, and real-time health checks[cite: 3].
- Configure automated alerting thresholds to notify engineering teams instantly and trigger automated failover scripts when anomalies are identified[cite: 3].